

SME CREDIT ANALYTICS

Loan Application & Portfolio Analytics Dashboard

End-to-end PostgreSQL, R, Statistics and R Shiny Portfolio Project

Project purpose

To simulate the work of a Credit-SME analytics team by transforming synthetic customer, application, processing-stage, loan and repayment data into validated management KPIs, statistical insights and an interactive R Shiny dashboard.

Technology Stack

PostgreSQL • SQL • R • dplyr • ggplot2 • Statistics • R Shiny • DT

Portfolio Project Report | October 2026

Dataset: Synthetic training data - no real customer or institutional data

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Report Outline

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How to read this report

The report focuses on business questions, validated analytical outputs and decision-support value. Detailed SQL and R scripts remain part of the project repository, while only representative outputs are included here.

Executive Summary

This project developed an end-to-end Credit-SME analytics workflow using a synthetic SME lending dataset. The solution combines relational data management in PostgreSQL, advanced SQL analysis, exploratory and statistical analysis in R, and an interactive R Shiny dashboard designed for management reporting and portfolio monitoring.

The analytical scope covers the full lending journey from application intake and processing stages through approval, disbursement, outstanding exposure and repayment performance. The project emphasizes reproducible KPI definitions, data-quality checks, operational turnaround-time analysis and portfolio-risk monitoring rather than predictive credit scoring, which is reserved for a future project phase.

Headline KPI	Validated Result	Management Meaning
Total applications	1,500	Overall SME financing demand in the synthetic portfolio
Approval rate	71.8%	Approved share of decided applications (Approved + Rejected)
Median turnaround time	15 days	Typical application decision time
Total disbursement	BDT 4.108 billion	Total value of disbursed SME facilities
Active outstanding	BDT 1.341 billion	Current non-closed portfolio exposure
PAR30	45.1%	Share of active outstanding associated with loans 30+ days past due
PAR90	12.35%	Share of active outstanding associated with loans 90+ days past due

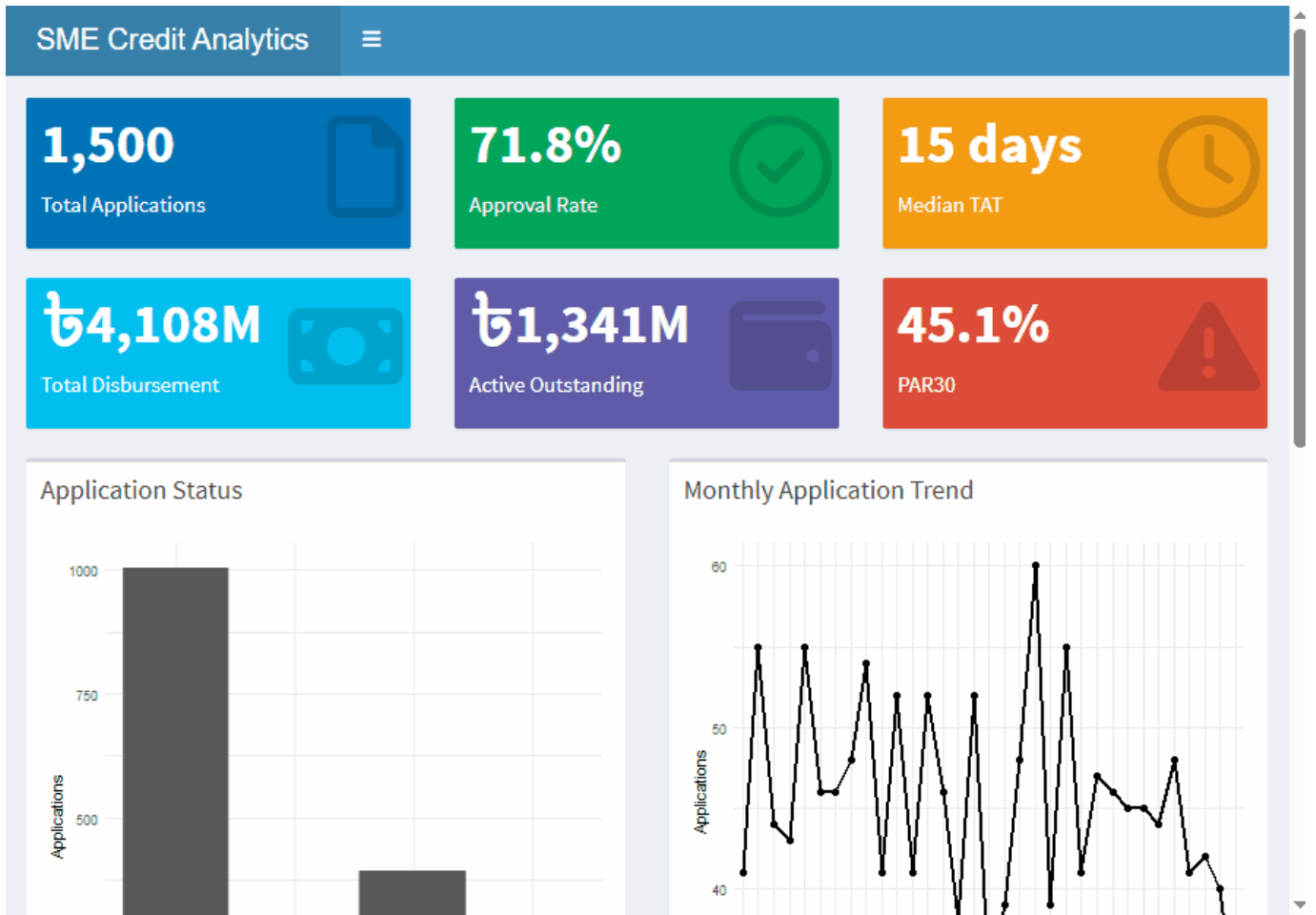


Figure 1. R Shiny Executive Overview showing the six headline management KPIs and application trend views.

The dashboard allows branch and business-sector filtering, enabling users to move from an overall portfolio view to more focused operational and risk analysis. The strongest findings include concentration of application volume and portfolio exposure in larger segments, materially elevated PAR30 in selected sectors and branches, a longer average duration in Credit Assessment than other processing stages, and a statistically meaningful difference in Debt-to-Revenue between approved and rejected applications in the synthetic dataset.

1. Project Background and Objectives

1.1 Business problem

A Credit-SME management team requires a consolidated view of loan demand, decision outcomes, application-processing efficiency, disbursement, current exposure and repayment deterioration. These measures are often stored across separate operational tables and therefore require relational integration, consistent KPI definitions and a reporting layer that supports drill-down analysis.

1.2 Project objective

The objective was to build a portfolio-ready analytics solution that converts raw SME lending records into management-level information. The project was designed to demonstrate practical proficiency in SQL, R, statistics, Shiny and credit-domain analytics within a single coherent workflow.

1.3 Analytical questions

- How many SME applications are received, approved, rejected, pending or withdrawn?
- How do application volume, approval rate and turnaround time vary across branches, sectors and time?
- Which processing stages contribute most to application turnaround time?
- How much has been disbursed and how much active exposure remains outstanding?
- Where is delinquency concentrated by DPD bucket, branch and business sector?
- Do borrower characteristics such as Debt-to-Revenue differ across application outcomes?
- Can management interactively filter the portfolio and observe the effect on all major KPIs?

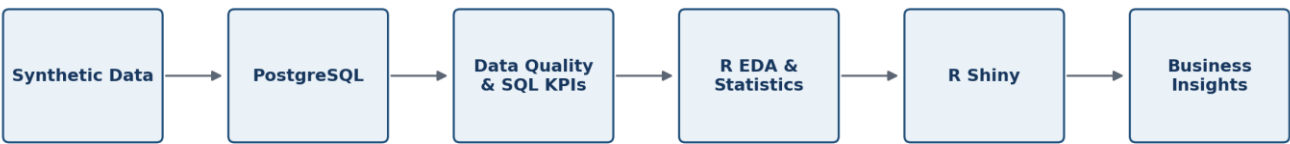


Figure 2. End-to-end analytical workflow used in the project.

2. Dataset and Relational Data Model

2.1 Dataset scope

The project uses fully synthetic data created for learning and portfolio demonstration. It does not represent real customers, a real financial institution or actual market outcomes. The application records span 1 January 2024 to 20 September 2026, while the lending and repayment tables extend through the associated portfolio-monitoring period.

Table	Rows	Grain / one row represents
Branches	8	One SME branch
Customers	1,000	One SME customer/business
Applications	1,500	One loan application
ApplicationStages	8,978	One application-stage event

Table	Rows	Grain / one row represents
Loans	995	One disbursed loan
Repayments	12,218	One scheduled repayment installment

Relational data model

One-to-many relationships preserve the analytical grain of each table.

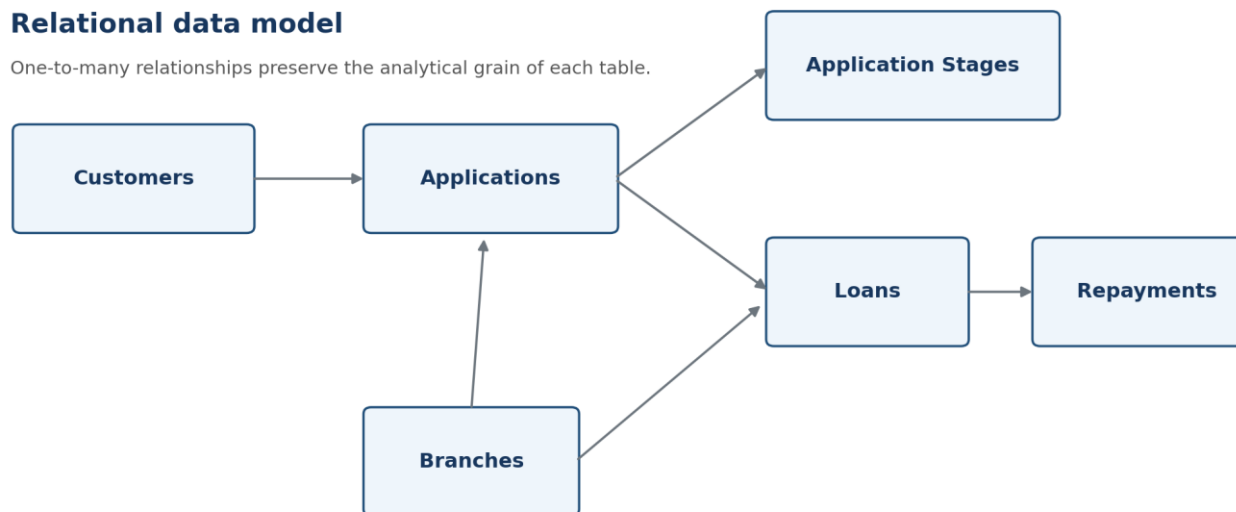


Figure 3. Relational data model used to connect customer, application, stage, loan and repayment information.

2.2 Importance of analytical grain

A key design principle throughout the analysis is to preserve the grain of each table. For example, joining one loan directly to many repayment rows can multiply loan-level balances and produce incorrect portfolio totals. Loan-level KPIs were therefore calculated from loan-grain data or from separately aggregated repayment results where appropriate.

3. Technology Stack and Analytical Methodology

Layer	Tools	Role in the solution
Database	PostgreSQL / pgAdmin	Relational storage, keys, joins, data-quality checks and KPI queries
Data analysis	R, dplyr, tidyr, lubridate	Data preparation, transformation and exploratory analysis
Visualization	ggplot2	Distribution, comparison and relationship plots
Statistics	R statistical functions	Confidence intervals, t-tests, chi-square, correlation and a logistic-regression preview
Dashboard	Shiny, shinydashboard, DT	Interactive management KPIs, charts, filters and tables

The solution intentionally separates database responsibilities from analytical and presentation responsibilities. PostgreSQL is used to define consistent relational logic and reporting metrics; R is used for flexible investigation and statistical analysis; Shiny converts the analytical layer into an interactive application.

4. Data Quality and Validation

Before calculating KPIs, the source tables were checked for uniqueness, completeness, referential integrity, valid dates and plausible financial values. This step is essential because technically correct SQL can still produce misleading results when the underlying records are inconsistent.

Validation check	Result	Interpretation
Duplicate customer IDs	0	Customer primary key is unique
Duplicate application IDs	0	Application primary key is unique
Missing sector / revenue / business age	0	Core customer profile fields are complete
Approved application without decision date	0	Decision logic is internally consistent
Decision date before application date	0	No impossible decision chronology detected
Negative requested/disbursed/outstanding values	0	No invalid negative financial balances found
Rejected applications with loans	0	No strong application-to-loan inconsistency detected
Approved applications without a loan	9	Flagged for follow-up; may represent approved but not yet disbursed cases
Outstanding balance greater than disbursement	0	Loan balances remain within simplified dataset rules
Negative DPD / negative repayment values	0	No invalid delinquency or repayment values found

Interpretation principle

Not every unusual record is an error. For example, an approved application without a loan may be waiting for documentation or disbursement. Data-quality flags should therefore trigger business review rather than automatic deletion.

5. SQL Analytics and Credit-SME KPI Framework

5.1 Application performance

Application outcomes show 1,004 approved, 394 rejected, 95 withdrawn and 7 pending applications. Using the project definition, the approval rate is calculated on decided applications only: $\text{Approved} / (\text{Approved} + \text{Rejected})$. This produces an approval rate of 71.8%. The overall application-status mix is shown below.

Application status	Count	Share of all applications
Approved	1,004	66.93%
Rejected	394	26.27%
Withdrawn	95	6.33%
Pending	7	0.47%

5.2 Turnaround time and processing stages

Across resolved applications, average turnaround time is 14.07 days, median turnaround time is 15 days and the 90th percentile is 19 days. The stage analysis shows that Credit Assessment has the longest average stage duration at 4.45 days, followed by Documentation at 3.53 days and Risk Review at 2.56 days.

Processing stage	Department	Average duration (days)
Credit Assessment	Credit	4.45
Documentation	Operations	3.53
Risk Review	Risk	2.56
Initial Screening	SME Sales	2.00
Approval Committee	Credit	1.97
Disbursement	Operations	1.48
Application Received	SME Sales	0.52

5.3 Portfolio exposure and delinquency

The portfolio contains 995 disbursed loans with total disbursement of approximately BDT 4.108 billion and average loan size of BDT 4.13 million. Active outstanding exposure is BDT 1.341 billion. PAR30 is 45.1% and PAR90 is 12.35% under the simplified project definitions.

DPD bucket	Active loans	Outstanding exposure
Current	116	BDT 345.7m
1-30 DPD	162	BDT 410.2m

DPD bucket	Active loans	Outstanding exposure
31-60 DPD	197	BDT 406.8m
61-90 DPD	47	BDT 99.8m
90+ DPD	54	BDT 78.7m

The project also applies CASE-based risk buckets, branch and sector PAR30 analysis, application aging, LAG-based month-over-month comparisons, three-month rolling averages, ROW_NUMBER for latest-stage identification, ranking functions and rule-based early-warning flags. These queries demonstrate how advanced SQL can move from descriptive reporting to operational monitoring.

6. Exploratory Data Analysis in R

6.1 Requested loan amounts

Requested loan amounts range from BDT 0.5 million to BDT 25 million. The mean request is BDT 4.72 million while the median is BDT 3.10 million, indicating a right-skewed distribution in which a relatively small number of large requests pull the average upward. For this reason, medians and distribution plots were used alongside means.

6.2 Branch turnaround-time distribution

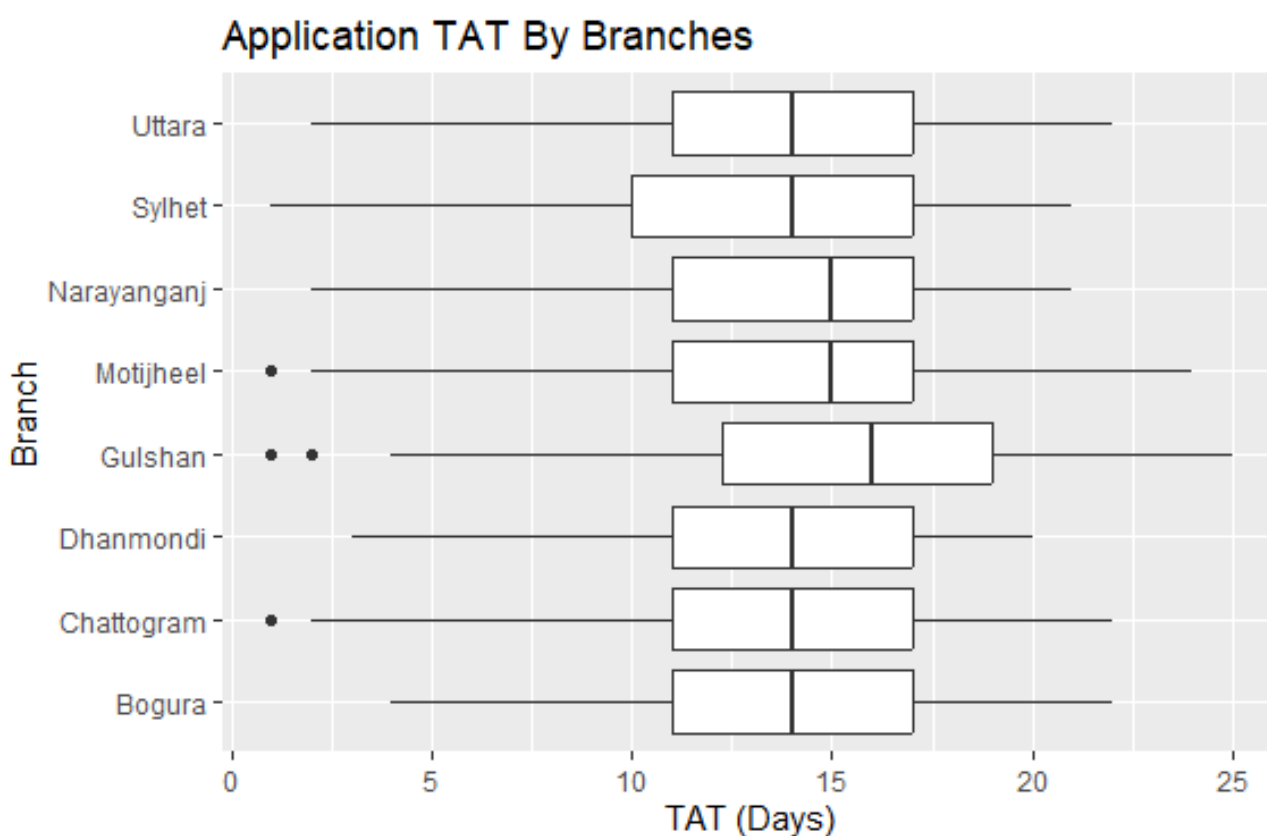


Figure 4. Application turnaround-time distributions by branch. Boxplots show both central tendency and variation.

Gulshan shows the highest median TAT among the branches in the dataset at 16 days, while most other branches have medians around 14-15 days. The distributional view is more informative than an average alone because it exposes inconsistent processing and extreme delays.

6.3 Portfolio risk by business sector

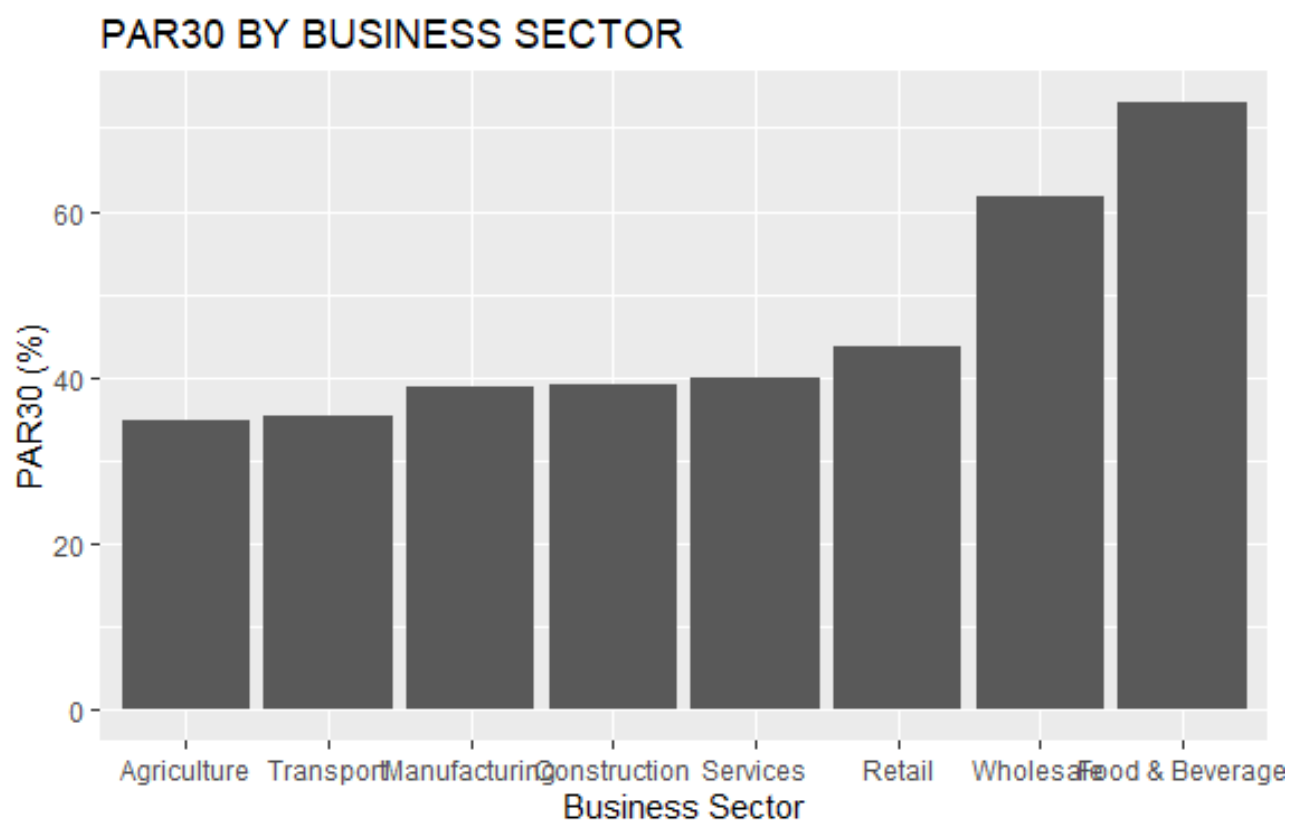


Figure 5. PAR30 by business sector for active loans.

Food & Beverage has the highest sector-level PAR30 at approximately 73.2%, followed by Wholesale at approximately 61.7%. Retail has the largest active outstanding exposure (about BDT 318.8 million) but a lower PAR30 of approximately 43.7%. This illustrates the difference between concentration risk (amount exposed) and delinquency rate (relative risk).

7. Statistical Analysis

Statistical analysis was used to move beyond descriptive differences and evaluate the strength of selected relationships. The analysis focused on confidence intervals, group comparisons, association tests, correlation and a simple univariate logistic-regression preview.

7.1 Approval-rate and TAT uncertainty

Estimate	Result
Approval rate	71.8%
Approx. 95% confidence interval for approval rate	69.4% to 74.1%
Mean TAT	14.07 days
95% confidence interval for mean TAT	13.85 to 14.30 days
Median TAT	15 days
TAT IQR	6 days

7.2 Debt-to-Revenue and application outcome

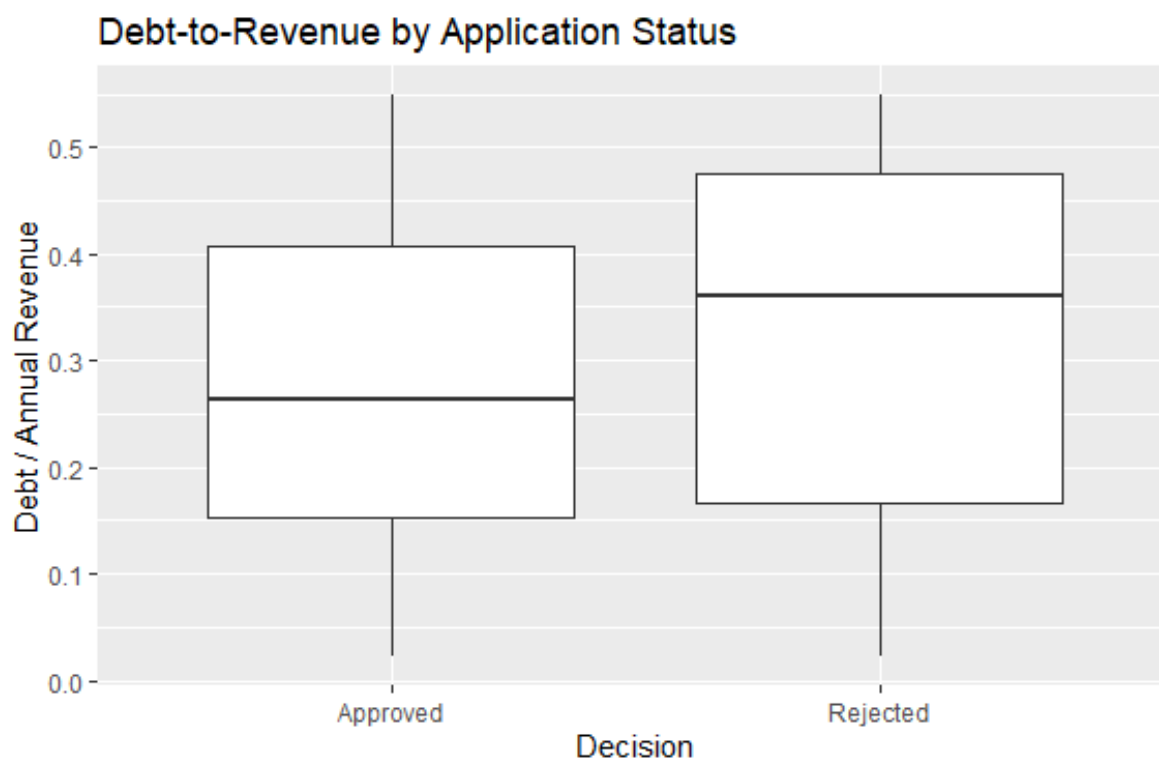


Figure 6. Debt-to-Revenue distribution for approved and rejected applications.

Approved applications have an average Debt-to-Revenue ratio of 27.5%, compared with 32.4% for rejected applications. A Welch two-sample t-test indicates a statistically detectable difference ($p < 0.001$), with the approved-minus-rejected mean difference of approximately -4.94 percentage points. This is an association within the synthetic dataset and should not be interpreted as causal evidence.

7.3 Sector and approval decision

A chi-square test of independence between business sector and Approved/Rejected decision does not show statistically strong evidence of an overall association in this synthetic dataset ($p \approx 0.415$). Although approval rates vary descriptively across sectors, the observed differences should not be overinterpreted without stronger evidence or multivariable analysis.

7.4 Revenue and requested financing

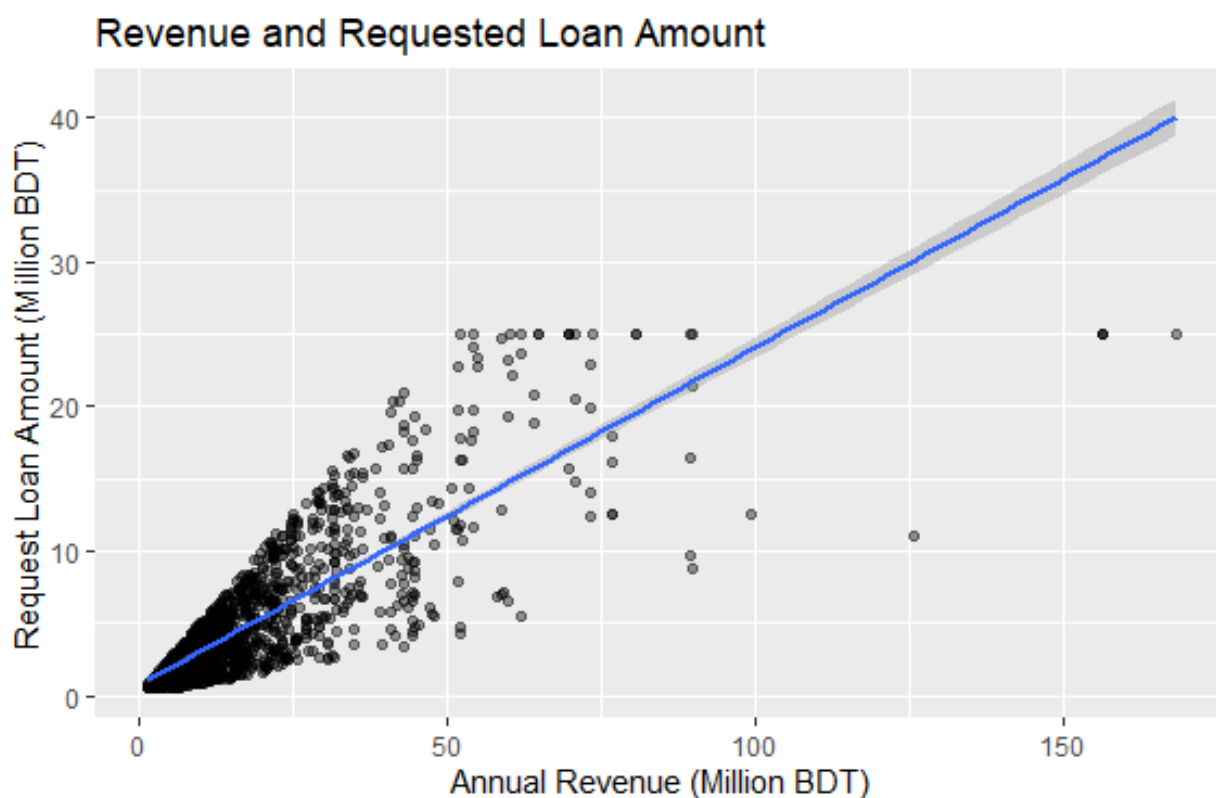


Figure 7. Strong positive relationship between annual revenue and requested loan amount.

Annual revenue and requested loan amount show a strong positive Pearson correlation ($r \approx 0.822$, $p < 0.001$). Larger businesses therefore tend to request larger facilities in the synthetic data. The scatterplot was used alongside the coefficient to check the direction, dispersion and presence of influential observations.

7.5 Logistic-regression preview

A simple logistic model using Debt-to-Revenue as the only predictor produces a negative coefficient for approval, consistent with the group comparison above. This model is included only as a statistical preview; it is not presented as a production credit-scoring model because it lacks multivariable controls, holdout validation, discrimination/calibration assessment and governance procedures.

8. R Shiny Management Dashboard

The Shiny application converts the analysis into an interactive decision-support tool. Branch and business-sector filters are global inputs; reactive datasets recalculate the KPI cards, charts and tables whenever the user changes a selection.

Reactive architecture

User selection → filtered application/loan data → KPI calculation → charts and tables → automatic dashboard refresh. This removes the need to rerun analysis manually for each branch or sector.

8.1 Executive Overview

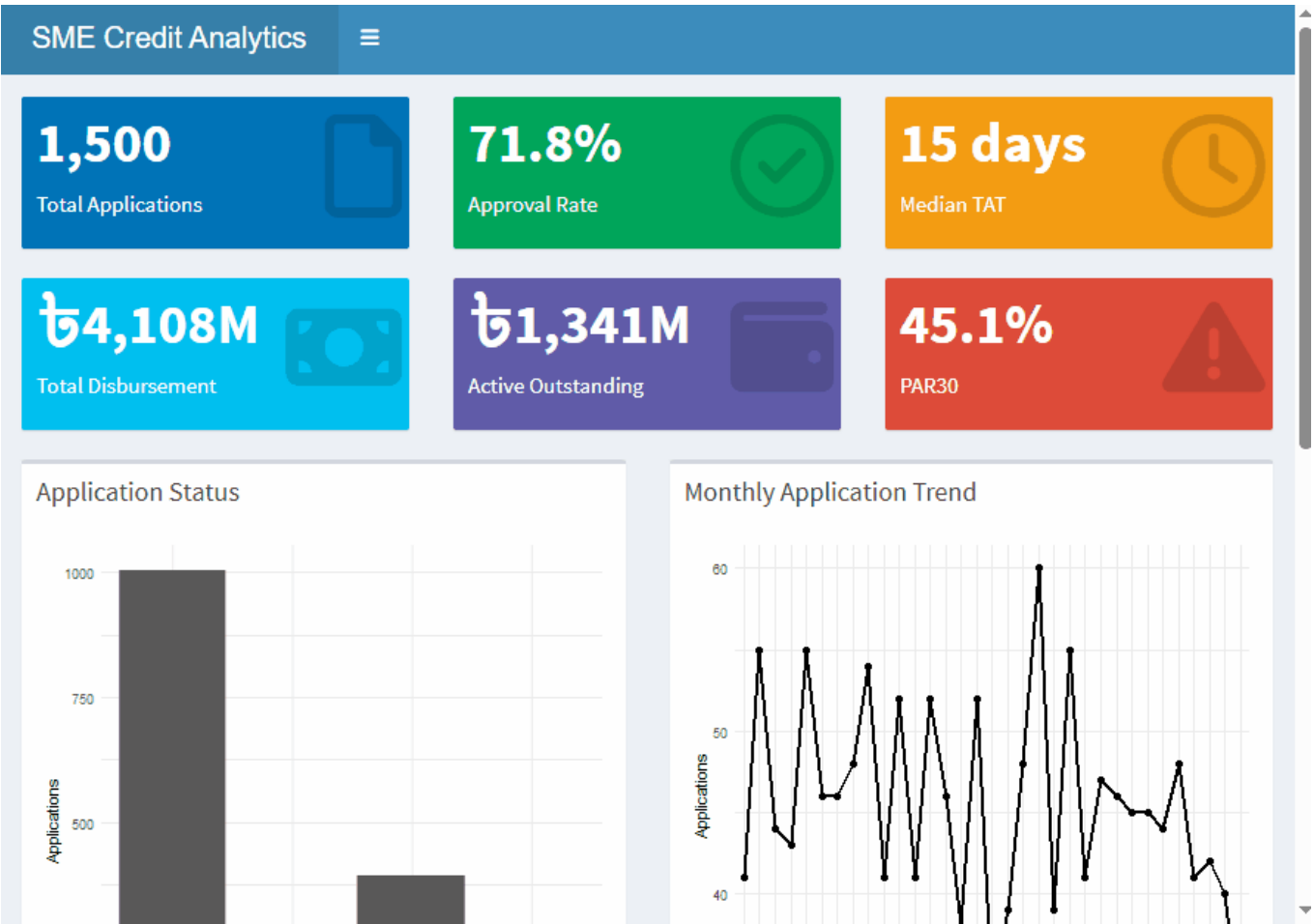


Figure 8. Executive Overview with portfolio-level KPIs and application trend charts.

8.2 Applications

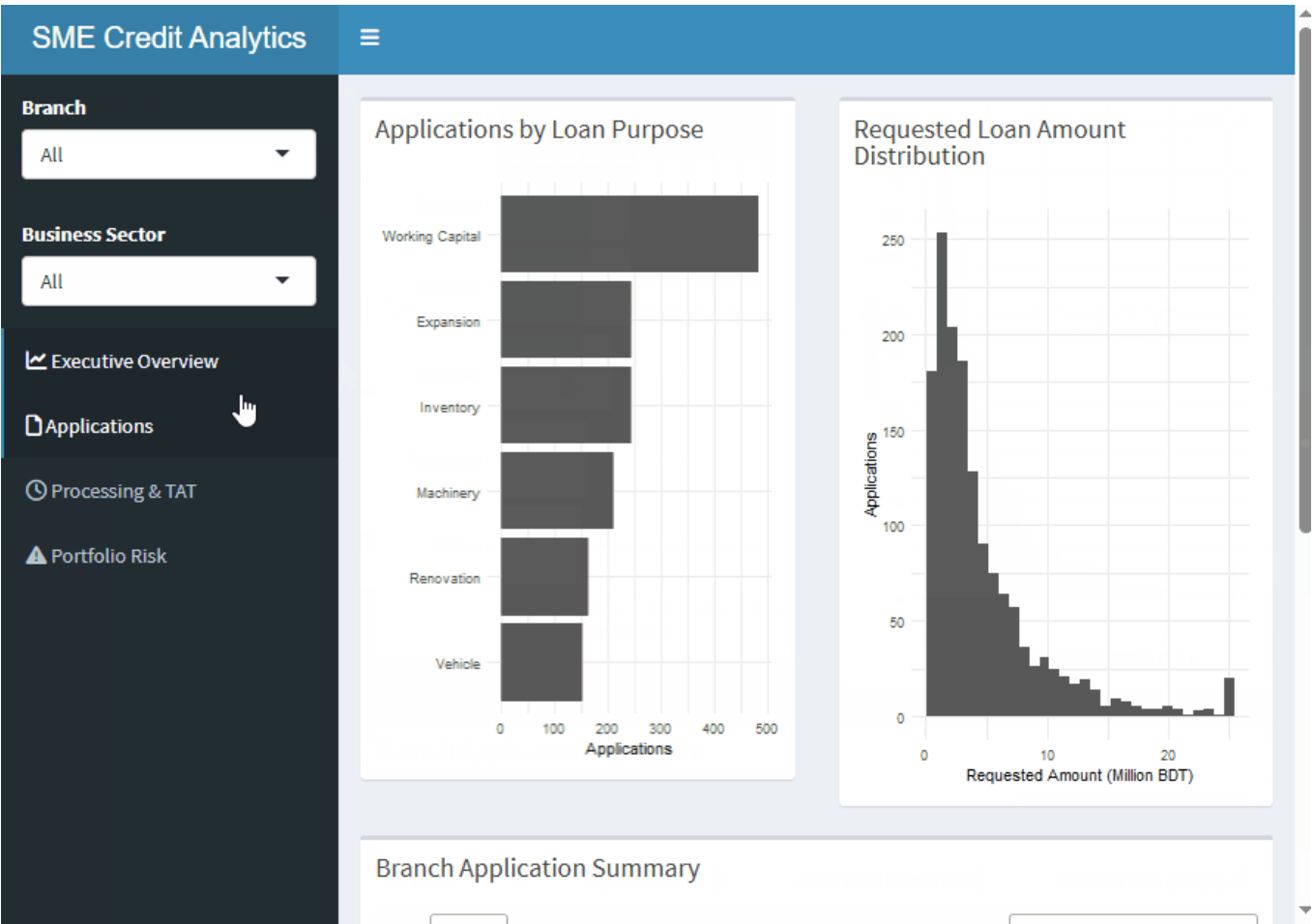


Figure 9. Applications page showing loan-purpose mix, requested-amount distribution and branch summary.

8.3 Processing & TAT

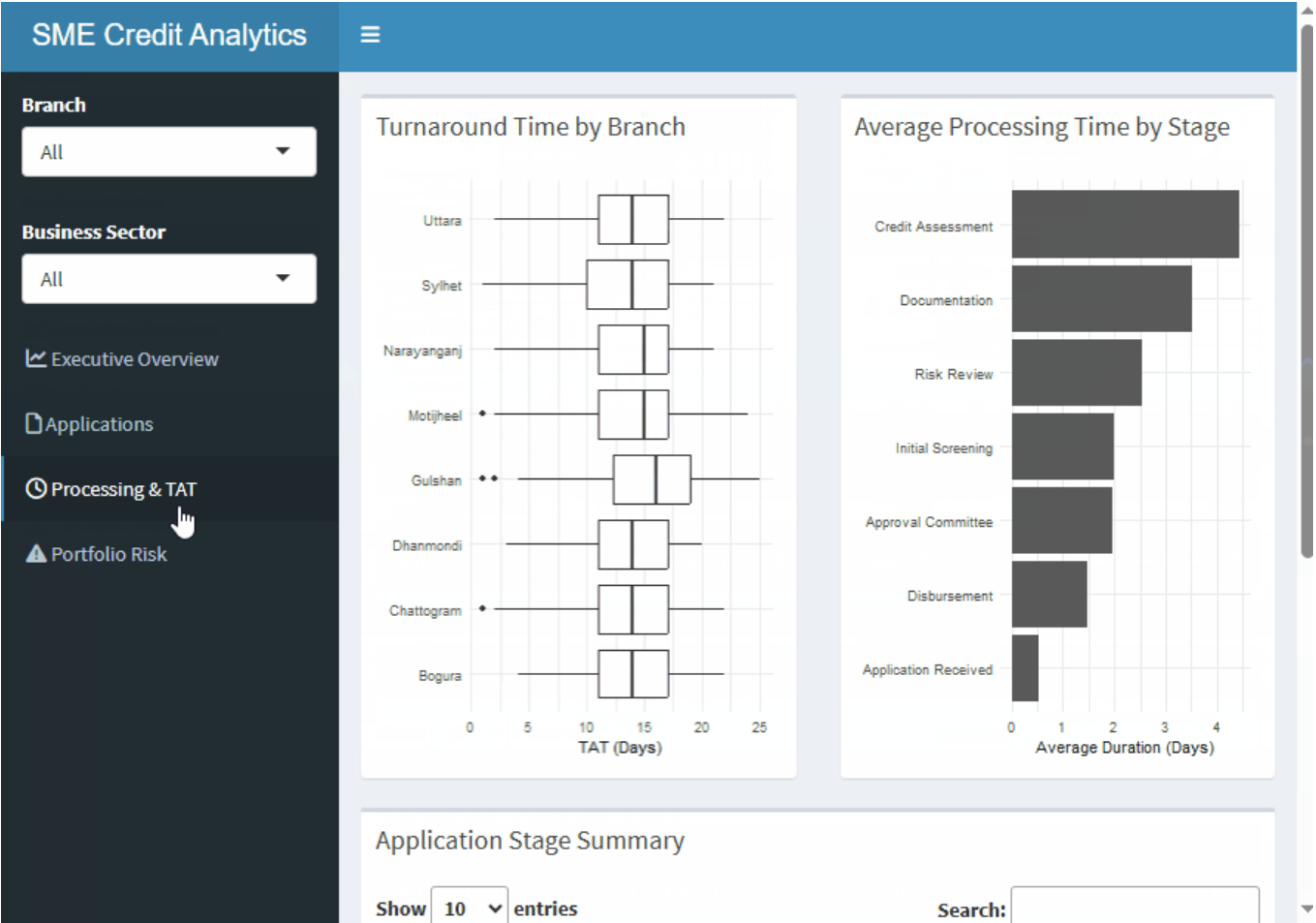


Figure 10. Processing & TAT page showing branch TAT distribution and average stage duration.

8.4 Portfolio Risk

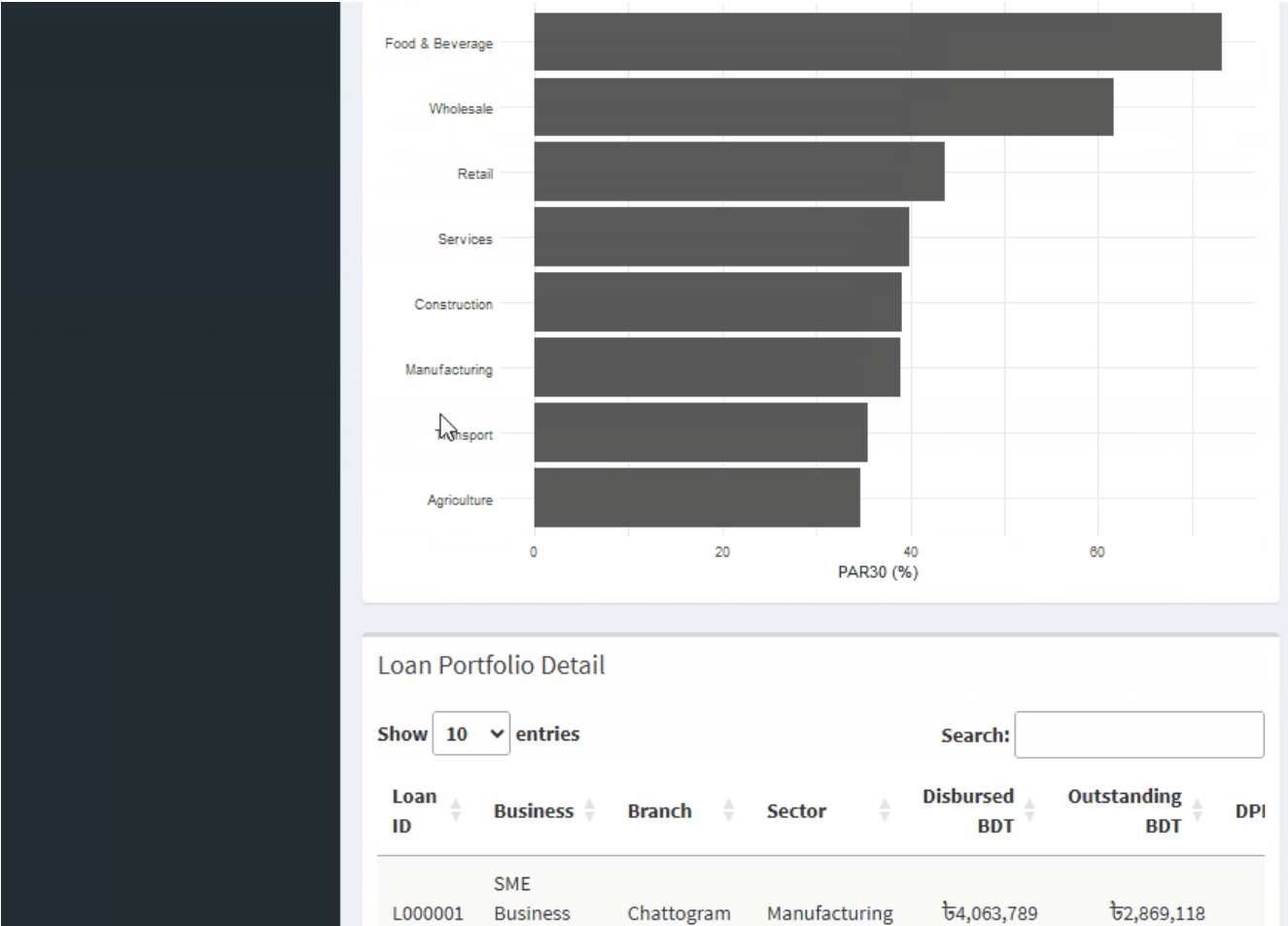


Figure 11. Portfolio Risk page showing PAR30 by business sector and loan-level detail.

The screen recording also confirms that selecting a branch or sector changes the KPI cards and charts. For example, choosing an individual branch produces a new application count, approval rate, TAT, disbursement, outstanding exposure and PAR30 without rerunning the R script. This demonstrates the core Shiny concept of reactivity.

9. Key Findings

Application outcomes

The dataset contains 1,500 applications; 1,004 are approved and 394 rejected, giving a 71.8% approval rate among decided applications.

Processing efficiency

Median TAT is 15 days. Credit Assessment is the longest average processing stage (4.45 days), making it the first workflow stage to examine when investigating delays.

Application concentration

Gulshan handles the highest application volume (316 applications), while Retail generates the largest sector application volume (345 applications). High volume should not automatically be interpreted as better branch or sector performance.

Portfolio concentration

Retail represents the largest active outstanding exposure at approximately BDT 318.8 million (23.8% of active outstanding), followed by Services at approximately BDT 295.8 million (22.1%).

Sector risk

Food & Beverage shows the highest sector PAR30 at approximately 73.2%, followed by Wholesale at approximately 61.7%. These figures warrant concentration and borrower-level investigation rather than immediate policy conclusions.

Branch risk

Bogura shows the highest branch PAR30 at approximately 65.9%, despite having one of the smaller active outstanding portfolios. Motijheel has the largest active outstanding amount among branches at approximately BDT 216.9 million with PAR30 of about 50.6%.

Borrower financial profile

Rejected applications show a higher average Debt-to-Revenue ratio than approved applications by approximately 4.94 percentage points; the difference is statistically detectable in the synthetic data.

Business size and financing demand

Annual revenue and requested financing are strongly positively correlated ($r \approx 0.822$), indicating that larger businesses generally request larger facilities in the synthetic portfolio.

10. Business Recommendations

- Prioritize stage-level TAT monitoring: Track Credit Assessment duration by branch and month because it has the longest average stage time. Use percentiles as well as averages so that severe delays are not hidden.
- Investigate high-PAR30 segments: Review Food & Beverage and Wholesale at borrower level to determine whether high PAR30 is driven by a few large facilities or broad-based deterioration. The same principle applies to Bogura at branch level.
- Monitor concentration and delinquency together: Use outstanding portfolio share and PAR30 simultaneously. A large exposure with moderate PAR30 may be economically more important than a small exposure with a high rate.
- Create an operational pending-applications watchlist: Use application aging and latest-stage logic to prioritize pending cases that have remained unusually long in a stage.
- Standardize KPI definitions: Maintain a single documented definition for approval rate, active outstanding, PAR30 and PAR90 so SQL, R and Shiny report the same figures.
- Extend monitoring toward early warning: Add repayment-migration analysis, roll rates and validated rule-based watchlist indicators before moving to predictive credit scoring.

11. Limitations and Quality Notes

- The dataset is synthetic. Results demonstrate analytical methods and should not be interpreted as actual SME-market evidence or institutional performance.
- The simplified PAR definitions are training definitions. Production institutions may use different exposure scopes, default definitions, restructuring rules or regulatory conventions.
- The statistical tests are primarily univariate and do not establish causality. Repeated applications from the same customer may also weaken strict independence assumptions.

- The logistic regression is exploratory only and was not validated as a predictive model.
- The project scripts were developed iteratively. For final publication, credentials should be moved to environment variables, package installation commands removed from runtime scripts, and all reporting dates standardized.
- Headline metrics in this report were independently recalculated from the underlying source tables before inclusion, using consistent active-portfolio and PAR definitions.

12. Future Development

The natural next step is a dedicated SME credit-risk modeling project. That work should preserve the data-quality and KPI discipline established here while adding predictive-model development and validation.

Future component	Planned capability
Credit scoring	Binary default target and borrower/application predictors
Modeling	Logistic regression plus benchmark machine-learning models
Model evaluation	ROC-AUC, KS, Gini, confusion matrix, calibration
Scorecard concepts	WoE / IV, risk grades and cutoff analysis
Monitoring	Population stability, drift, approval/default performance and model governance
Dashboard extension	Model score distribution, risk bands and early-warning watchlist

13. Conclusion

This project demonstrates an end-to-end analytics workflow for SME lending. PostgreSQL was used to structure relational data, validate quality and calculate management KPIs; R supported exploratory and statistical analysis; and Shiny transformed the results into an interactive dashboard. The project therefore demonstrates not only technical skills in SQL and R, but also the ability to define credit-domain metrics, preserve analytical grain, distinguish observation from causation and communicate findings in a management-oriented format.

As a portfolio project, the strongest value is the integration of database engineering, analytical reasoning, statistics and dashboard delivery around a realistic Credit-SME business process. The solution is intentionally transparent and reproducible, providing a strong foundation for the next project on credit scoring and default prediction.

Appendix A - KPI Definitions

KPI	Project definition
Approval Rate	$\text{Approved} / (\text{Approved} + \text{Rejected}) \times 100$
Rejection Rate	$\text{Rejected} / (\text{Approved} + \text{Rejected}) \times 100$
Pending Rate	$\text{Pending} / \text{Total Applications} \times 100$
TAT	Decision Date - Application Date
Active Outstanding	Sum of outstanding balance for loans whose status is not Closed
PAR30	$\text{Outstanding balance of active loans with DPD} \geq 30 / \text{active outstanding} \times 100$
PAR90	$\text{Outstanding balance of active loans with DPD} \geq 90 / \text{active outstanding} \times 100$
Portfolio Share	$\text{Sector active outstanding} / \text{total active outstanding} \times 100$
Debt-to-Revenue	Existing Debt / Annual Revenue
Collateral Coverage	Collateral Value / Requested Amount

Appendix B - Suggested Repository Structure

```
credit-sme-loan-analytics/  
├── README.md  
├── app.R  
├── data/  
│   └── data_dictionary.md  
├── sql/  
│   ├── 01_create_tables.sql  
│   ├── 02_data_quality.sql  
│   ├── 03_business_analysis.sql  
│   ├── 04_kpi_framework.sql  
│   └── 05_advanced_analysis.sql  
├── analysis/  
│   ├── exploratory_data_analysis.R  
│   └── statistical_analysis.R  
├── screenshots/  
│   ├── executive_overview.png  
│   ├── applications.png  
│   ├── processing_tat.png  
│   └── portfolio_risk.png  
└── docs/  
    └── project_report.pdf
```

Portfolio statement

This project uses synthetic data only. No real customer, employer or institutional confidential information is included in the analytical dataset.